

Housing Character Edge Weights via American Community Survey

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Abstract

Housing stock character—the mix of single-family and multifamily units, ownership versus rental tenure, and the vintage of the built environment—encodes information about community structure that is absent from occupational, economic, or administrative data. Single-family homeowner neighborhoods share zoning politics, property tax interests, school district bond capacities, HOA governance structures, and a common stake in neighborhood stability that is fundamentally different from the interests of high-rise rental corridors or postwar apartment complexes. Yet no existing redistricting algorithm operationalizes this distinction.

This paper introduces M.3, the housing character implementation of the M-track community character framework [Della-Libera, 2025]. Using three American Community Survey (ACS) 5-year tables—B25024 (units in structure), B25003 (tenure by occupancy status), and B25035 (median year structure built)—we construct a four-component housing character vector for each census tract: percentage single-family units, percentage large-multifamily units, owner-occupancy rate, and housing vintage score. The cosine similarity of adjacent tracts’ housing character vectors modifies METIS edge weights via the standard M-track blend formula.

M.3 has a distinctive implementation advantage: the ACS tables it requires are available through the same Census API infrastructure already used by `bisect fetch`. The implemented `acs-housing` fetch path writes the four derived columns, and `-weights-override housing-character` applies the edge-weight blend.

Implementation mechanics are backed by a hash-bound smoke package; empirical outcome results remain future work until full plan runs are archived. The empirical design targets North Carolina (14 congressional districts, 2020 vintage): expected outcomes include within-district homogenization of housing type and clustering of pre-1960 historic neighborhoods. The legal foundation draws on neighborhood stability

jurisprudence, historic preservation district doctrine, and long-tenured homeowner representation interests recognized in state redistricting criteria.

Keywords

Housing character, Communities of interest, American Community Survey, ACS B25024, Edge weights, METIS, Cosine similarity, Homeownership, Housing vintage, Algorithmic redistricting, NIMBYism, Neighborhood stability

1 Introduction

At a 2021 public hearing of North Carolina’s redistricting commission, a resident of Raleigh’s Cameron Park neighborhood testified that splitting her historic district across two congressional seats would deprive the neighborhood’s long-tenured homeowners of a coherent representative voice on federal preservation funding, housing finance policy, and historic tax credit renewal. Her testimony was noted but could not be quantified. The commission had no tool that could measure whether Cameron Park and the adjacent Five Points neighborhood shared a community bond that the Oakwood tract across the freeway did not.

This paper provides that measurement.

The M-track community character framework, introduced in M.0 [Della-Libera, 2025], operationalizes diverse community character signals as edge weights in the census-tract adjacency graph used by BISECT. Tracts whose character vectors are similar receive heavy edges; tracts with dissimilar character receive light edges. METIS’s minimum-cut objective then respects community structure automatically: cuts run along light edges (natural community transitions) and avoid heavy edges (strong community bonds).

M.3 implements the housing character signal. It is the third M-track implementation paper, addressing a community character dimension that neither M.1 (economic character, workplace-side LODS data) nor M.6 (administrative zone co-membership) capture: the residential built environment. How old is the housing stock? Is it predominantly detached single-family, attached rowhouses, or large multifamily buildings? Do residents own or rent? These three questions—answered by three ACS tables—define a housing character vector that separates single-family homeowner suburbs from apartment corridors, postwar subdivisions from pre-war rowhouse neighborhoods, and owner-occupied historic districts from transient rental enclaves.

Why housing type predicts community. The connection between housing type and community structure is not sociological speculation; it is encoded in law. A single-family homeowner is, by definition, a property taxpayer with a direct financial stake in local school quality, infrastructure maintenance, and zoning stability. A multifamily renter in the same tract shares none of these fiscal bonds. The two populations have different relationships to local government, different sensitivities to federal housing finance policy, different interests in historic tax credit programs, and different vulnerabilities to local zoning variance decisions [Fischel, 2001]. They may be geographically contiguous—a rowhouse block and an adjacent apartment tower may literally share a property line—but they are not members of the same community of interest in the legally relevant sense.

The NIMBYism signal. One empirically documented mechanism through which housing character becomes a community signal is the politics of land-use opposition, commonly abbreviated as NIMBYism (Not In My Back Yard). Communities that are predominantly single-family and owner-occupied share a strong revealed preference for maintaining the existing housing stock character: they oppose upzoning, density increases, and new multifamily permitting at high rates [Fischel, 2001, Einstein et al., 2019]. This shared political preference is partisan-neutral—NIMBYism is documented in liberal San Francisco and conservative Houston alike—but it is a strong predictor of community cohesion and shared political interest. Tracts with similar housing character (both highly single-family-and-owner) share political interests that cross-character tracts (one single-family, one large multifamily) do not.

Contribution. This paper makes the following contributions:

1. **Four-component housing character vector** derived from three ACS tables (B25024, B25003, B25035), with a precise formula for each component and a rationale grounded in community structure theory (§3).
2. **Zero-cost implementation path:** M.3 requires no new data download infrastructure; it extends the existing ACS demographics CSV with four additional columns (§3).
3. **Empirical design:** a pre-specified experimental protocol for North Carolina, with falsifiable predictions about within-district housing homogeneity and historic neighborhood clustering (§4).
4. **Legal grounding:** a practitioner-oriented analysis of neighborhood stability doctrine, historic preservation district law, and the homeowner representation interest (§5).

Paper organization. Section 2 reviews the ACS methodology and the political science literature linking housing type to community. Section 3 defines the four-component character vector and the blend formula. Section 4 describes the implementation evidence boundary and future empirical design. Section 5 provides legal grounding. Section 6 concludes.

2 Background

2.1 ACS 5-Year Estimates: Methodology and Reliability

The American Community Survey (ACS) is a continuous demographic survey conducted by the U.S. Census Bureau, collecting information from approximately 3.5 million housing units annually [U.S. Census Bureau, 2020]. The 5-year estimates aggregate five calendar years of data to produce reliable tract-level estimates; single-year ACS estimates are not published at the census tract level because the sample is too small. The 2016–2020 five-year estimates (“2020 ACS 5-year”) represent the most recent tract-level housing data available as of this writing and correspond to the 2020 census tract boundaries used by BISECT’s redistricting pipeline.

Margin of error. Because ACS estimates are survey-based, each published figure carries a margin of error (MOE) at the 90% confidence interval. For housing-stock variables at the census tract level, MOEs are typically within 10–15% of the estimate for tracts with population above 2,000. Smaller tracts and tracts in rural areas may have larger relative MOEs. For the purposes of the housing character vector, we treat the published point estimates as the true values and note that MOE-based weighting of the similarity metric is a potential extension. Formula and edge-weight mechanics are implementation-backed; empirical outcome claims remain future work until state plan runs are archived.

Tables used. Table 1 lists the three ACS tables from which the housing character components are derived. All three are available at census tract geography for every state, published in the ACS 5-year series.

Table 1: ACS 5-year tables used for housing character vector construction.

Table	Name	Housing character component
B25024	Units in Structure	Single-family fraction, multifamily fraction
B25003	Tenure	Owner-occupancy rate
B25035	Median Year Structure Built	Housing vintage score

2.2 Why Housing Type Predicts Community Structure

2.2.1 Homeownership and Civic Participation

The relationship between homeownership and civic engagement is one of the most studied questions in American political sociology. DiPasquale and Glaeser [1999] find in a national panel that homeowners are significantly more likely to know their congressional representative, participate in local organizations, and vote in local elections than otherwise-identical renters. The authors attribute the effect to the financial stake homeowners hold in local government outcomes: school quality, property tax rates, zoning variances, and infrastructure maintenance all directly affect property values in a way they do not affect rental costs.

Fischel [2001] elaborates this mechanism through the concept of the “homevoter hypothesis”: homeowners, unlike renters, cannot diversify their housing investment, so they vote as if their home equity were their primary financial asset. They vote for school board candidates, against upzoning proposals, and for infrastructure maintenance bonds with a consistency that non-homeowners do not exhibit. The political implication for redistricting is direct: a tract that is predominantly owner-occupied has a recognizably different political interest profile from an adjacent tract that is predominantly rental. They are not members of the same community of interest on housing and land-use policy.

2.2.2 Neighborhood Stability and the Filtering Theory

The lifecycle of the housing stock—its vintage—is another dimension of community structure. The “filtering” theory of housing markets [Rosenthal, 2014] holds that new construction at the high end of the market gradually filters down to lower income occupants as the housing ages. A tract built before 1940 and still predominantly owner-occupied is a community that has resisted the filtering dynamic: it has maintained its character across multiple generations of residents. Historic preservation districts, which are legally recognized institutional forms of this resistance, are typically composed of pre-1940 housing stock.

The community stability of pre-1960 neighborhoods is not merely a function of the age of their buildings; it is a function of the long tenures of their residents [Weaver, 2020]. Long-tenured homeowners have different representation interests from recent arrivals: they are invested in the long-run trajectory of local institutions, they have organized around neighborhood associations with multi-decade histories, and they have fought and won (or lost) land-use battles that shaped the current built environment. Splitting such a neighborhood across congressional district boundaries fragments a community of interest in the most literal sense.

2.2.3 NIMBYism as a Community Signal

Einstein, Palmer, and Glick [2019] analyze 2015–2017 local planning board meeting minutes from Massachusetts and find that participants who speak against housing development proposals are disproportionately homeowners, older, and white, while the broader community they claim to represent is more diverse and more renting. Their finding—that homeowner communities systematically mobilize against density increases—documents the political homogeneity of single-family homeowner communities. This homogeneity is precisely what the housing character vector is designed to measure: tracts where the homeowner fraction and single-family fraction are both high share a political interest profile that low-SF, high-rental tracts do not.

The NIMBYism signal is partisan-neutral in origin. Single-family homeowner communities with strong opposition to upzoning are documented in liberal San Francisco [Einstein et al., 2019], moderate suburban New Jersey [Fischel, 2001], and conservative Houston [NOLO, 2022]. The housing character signal does not predict partisan vote share; it predicts community cohesion on the housing-and-land-use dimension of political interest.

2.2.4 Implementation Position in the M-Track Framework

The M-track framework (M.0) defines a family of community character signals from M.1 (economic character) to M.7 (transit accessibility). Table 2 places M.3 within that family. M.3 occupies the residential built environment dimension, which is not covered by any other M-track signal. M.1 covers the workplace-side economic character; M.3 covers the residential-side physical character. M.6 covers administrative zone co-membership, which overlaps with M.3 in the school district dimension (school district membership correlates strongly with single-family homeowner neighborhoods) but is legally and computationally distinct.

Table 2: Position of M.3 in the M-track signal taxonomy.

Paper	Signal	Data source
M.1	Economic character (workplace)	LODES WAC
M.2	Land use character	NLCD
M.3	Housing character (residential)	ACS B25024/B25003/B25035
M.4	Commuting shed	LODES OD
M.5	Topographic character	USGS 3DEP
M.6	Administrative zone co-membership	TIGER/Line special districts
M.7	Transit accessibility	NTD / GTFS

3 Housing Character Algorithm

3.1 ACS Table Structure

B25024 — Units in Structure. Table B25024 reports the number of housing units in each census tract broken down by structural type. The categories are: 1-unit detached (B25024_002), 1-unit attached (B25024_003), 2 units (B25024_004), 3 or 4 units (B25024_005), 5 to 9 units (B25024_006), 10 to 19 units (B25024_007), 20 to 49 units (B25024_008), 50 or more units (B25024_009), mobile home (B25024_010), boat, RV, van, etc. (B25024_011). The universe is all housing units in the tract. The total is B25024_001.

B25003 — Tenure. Table B25003 reports the number of occupied housing units by tenure: owner-occupied (B25003_002) and renter-occupied (B25003_003). The universe is occupied housing units. The total is B25003_001.

B25035 — Median Year Structure Built. Table B25035 reports the median calendar year in which the housing structures in the tract were built, as a single integer. This is a tract-level median, not a microdata file. The universe is all housing units.

3.2 Four-Component Housing Character Vector

For each census tract t , we define the housing character vector

$$\mathbf{h}(t) = (\text{SF}(t), \text{MF}(t), \text{OO}(t), \text{VT}(t)) \in [0, 1]^4, \quad (1)$$

where the four components are defined as follows.

Single-family fraction (SF).

$$\text{SF}(t) = \frac{\text{B25024_002}(t) + \text{B25024_003}(t)}{\text{B25024_001}(t)}, \quad (2)$$

the fraction of units in 1-unit detached or attached structures. $\text{SF}(t) \in [0, 1]$. High values (> 0.80) characterize suburban single-family neighborhoods and urban rowhouse districts. Low values (< 0.20) characterize high-rise apartment corridors and dense urban cores.

Large-multifamily fraction (MF).

$$\text{MF}(t) = \frac{\text{B25024_007}(t) + \text{B25024_008}(t) + \text{B25024_009}(t)}{\text{B25024_001}(t)}, \quad (3)$$

the fraction of units in structures with 10 or more units. Note: the M.4 spec uses “5+ units” informally; the precise B25024 category boundary for large multifamily is 10 units (B25024_007: 10–19 units; B25024_008: 20–49 units; B25024_009: 50 or more units). Structures with 5–9 units (B25024_006) are excluded from the large-multifamily fraction because they are more similar in community character to small apartment buildings than to large elevator buildings. $MF(t) \in [0, 1]$. High values (> 0.50) characterize apartment corridors, urban high-density residential zones, and transit-oriented development nodes. The MF component distinguishes apartment-corridor communities from small-multifamily neighborhoods (duplexes, triplexes) that may share more community characteristics with single-family areas.

Owner-occupancy rate (OO).

$$OO(t) = \frac{B25003_002(t)}{B25003_001(t)}, \quad (4)$$

the fraction of occupied units that are owner-occupied. $OO(t) \in [0, 1]$. High values (> 0.70) characterize established homeowner communities. Low values (< 0.30) characterize predominantly rental communities such as university neighborhoods, downtown cores, and transit-oriented apartment districts.

Housing vintage score (VT).

$$VT(t) = \max\left(0, \min\left(1, 1.0 - \frac{B25035(t) - 1940}{2020 - 1940}\right)\right), \quad (5)$$

a score that maps the median year built to the interval $[0, 1]$, where higher values indicate older housing stock. *Implementation note:* The M.3 spec approximates this as $(2024 - B25035)/100$; the present formula supersedes that approximation. The spec formula is unbounded (it exceeds 1.0 for tracts with median year before 1924) and uses an arbitrary divisor. Equation (5) is the implementation target: it is clamped to $[0, 1]$, calibrated on the 1940–2020 housing stock range, and consistent with the implementation test invariants. The formula is calibrated so that:

- Median year 1940 or earlier: $VT = 1.0$ (fully historic).
- Median year 1980: $VT = 0.5$ (mid-century suburban).
- Median year 2010: $VT = 0.125$ (new suburban).

- Median year 2020 or later: $VT = 0.0$ (new construction, clamped).

The $\max(0, \min(1, \cdot))$ clamp handles the edge cases of tracts with B25035 values outside the 1940–2020 calibration range.

3.3 Cosine Similarity and Edge Weight

The housing character similarity between adjacent tracts u and v is the standard cosine similarity of their housing character vectors:

$$\text{sim}_h(u, v) = \frac{\mathbf{h}(u) \cdot \mathbf{h}(v)}{\|\mathbf{h}(u)\| \|\mathbf{h}(v)\|} \in [0, 1], \quad (6)$$

which is well-defined when neither vector is the zero vector. Because all four components are non-negative by construction, the cosine similarity is always in $[0, 1]$.

Zero-vector handling. A census tract can produce a zero housing character vector only if all four of B25024_001, B25003_001, and B25035 are zero or missing—in practice, a tract with no housing units (e.g., an industrial or parkland tract with residential census population assigned to it). Such tracts receive the conservative substitution $\text{sim}_h = 1.0$ (treat as perfectly similar to any neighbor), leaving the edge weight governed entirely by the geographic base weight.

Edge weight blend formula. Following the M-track blend convention established in M.1 and M.9, the housing-character edge weight is:

$$w_h(u, v) = w_{\text{geo}}(u, v) \times (\alpha + (1 - \alpha) \text{sim}_h(u, v)), \quad (7)$$

with default $\alpha = 0.5$. At $\alpha = 0.5$ and $\text{sim}_h = 1.0$ (identical housing characters), the weight equals w_{geo} : maximum community bond. At $\alpha = 0.5$ and $\text{sim}_h = 0.0$ (orthogonal housing characters), the weight equals $0.5 \times w_{\text{geo}}$: the geographic bond is halved rather than zeroed, preventing free-cut instability in METIS.

3.4 Prototypical Housing Character Vectors

Table 3 gives three prototypical tract types and their housing character vectors to build intuition for the cosine similarity geometry.

The cosine similarity matrix reveals the community geometry.

The similarity matrix encodes a central finding of the housing character framework:

Table 3: Prototypical housing character vectors for three tract types. SF = single-family fraction, MF = large-multifamily fraction, OO = owner-occupancy rate, VT = vintage score.

Type	Example	SF	MF	OO	VT
Single-family suburb	Cary, NC residential	0.88	0.02	0.75	0.30
Historic rowhouse district	Raleigh Cameron Park	0.72	0.05	0.65	0.85
Apartment corridor	Charlotte South End	0.08	0.72	0.18	0.20

Table 4: Cosine similarity between prototypical housing character tract types.

	Single-family suburb	Historic rowhouse	Apartment corridor
Single-family suburb	1.00	0.94	0.11
Historic rowhouse	0.94	1.00	0.20
Apartment corridor	0.11	0.20	1.00

- **Single-family suburb to historic rowhouse (0.94).** Both are predominantly single-family, owner-occupied communities. Despite the vintage difference, they share housing character. METIS should strongly avoid cutting the edge between them.
- **Suburb or rowhouse to apartment corridor (<0.20).** The housing character community boundary is the sharpest possible cut seam: single-family owner versus large-multifamily renter. METIS is invited to place the district boundary here.
- **Apartment corridor to apartment corridor (1.00 for identical vectors).** Apartment corridor communities belong together: they share rental market dynamics, transit-oriented mobility patterns, and land-use policy interests.

3.5 No New Download Infrastructure Required

A key practical advantage of M.3 is that the three ACS tables (B25024, B25003, B25035) are available from the same Census API endpoint already used by `bisect fetch` to populate the demographics CSV. The current ACS fetch retrieves tables covering race, age, income, and educational attainment at the census tract level. Adding B25024, B25003, and B25035 now uses the existing `bisect fetch -type acs-housing` path:

1. The fetch list requests ACS B25024, B25003, and B25035 tract variables from the standard Census ACS API endpoint.
2. The derived CSV contains four output columns: `pct_single_family`, `pct_multifamily`, `pct_owner`, and `housing_vintage`.

3. Implementing the four formulas (Equations 2–5) in the ACS housing preprocessing step.
4. Applying the cosine similarity and blend formula (Equations 6 and 7) in the **housing-character** weight mode.

No new API calls to new endpoints, no new file formats, and no changes to the adjacency graph pipeline are required. The CLI flag is **-weights-override housing-character**.

The current implementation treats B25024 large multifamily as 10+ unit structures (B25024_007E–B25024_009E) and uses B25024_001E as the denominator. Mobile homes and boat/RV/van units are part of the denominator but are not counted as large multifamily.

3.6 Test Invariants (L0)

The following invariants must hold on every build:

single_family_tracts_high_similarity For two tracts with $SF > 0.85$, $OO > 0.70$, and $|VT_u - VT_v| < 0.20$: $\text{sim}_h(u, v) > 0.85$.

mixed_tenure_vs_homogeneous_low_similarity For u with $OO = 0.9$, $SF = 0.9$ and v with $OO = 0.1$, $MF = 0.8$: $\text{sim}_h(u, v) < 0.30$.

vintage_far_apart_low_similarity For u with $VT = 0.10$ (built ≈ 2012) and v with $VT = 0.80$ (built ≈ 1956), holding SF and OO equal: $\text{sim}_h(u, v) < 0.50$.

similarity_symmetric $\text{sim}_h(u, v) = \text{sim}_h(v, u)$ for all tract pairs.

weight_stays_nonneg $w_h(u, v) \geq 0$ for all edges after blending.

The repository also includes the hash-bound smoke package **docs/examples/m-community-character-**. It verifies the ACS housing formulas, denominator defaults, and edge-weight blend through **bisect-cli::housing_evidence**.

4 Implementation and Evidence Boundary

The implementation slice for M.3 is now present in BISECT. The **acs-housing** fetch path derives the four housing-character columns, and the **housing-character** weight mode applies the cosine-similarity blend to graph edges. This section separates implemented evidence from future empirical claims: the formula and edge-weight mechanics are package verified, while state-level outcome results remain future work until full NC-14 runs are archived.

4.1 Target State and Configuration

The primary empirical target is North Carolina, 14 congressional districts, 2020 census vintage. North Carolina provides an ideal test case because it contains all three prototypical housing character communities within a single state:

- **Historic neighborhoods.** Raleigh’s Cameron Park, Five Points, and Oakwood; Durham’s Trinity Park and Watts-Hillandale; Charlotte’s Dilworth and Myers Park. These neighborhoods have median year built predominantly before 1950, high owner-occupancy, and predominantly single-family or attached rowhouse stock.
- **Postwar single-family suburbs.** Cary, Apex, Morrisville, and the Research Triangle Park periphery. Median year built 1970–2000, high owner-occupancy, predominantly detached single-family.
- **Apartment corridors.** Charlotte’s South End and South Park; Raleigh’s North Hills and Glenwood South; Durham’s downtown apartment district. Predominantly large multifamily, high rental fraction, recent construction.

The baseline comparison is `-weights-override geographic` on the same NC-14 plan configuration (convergence search, $T = 600$, balance tolerance = 0.5%).

4.2 Primary Empirical Hypothesis

H1: Within-district housing homogeneity improves. The M-track primary metric for M.3 is the mean within-district standard deviation of `pct_single_family` across all 14 NC districts. Formally:

$$\bar{\sigma}_{\text{SF}} = \frac{1}{14} \sum_{i=1}^{14} \text{std}\{\text{SF}(t) : t \in D_i\}. \quad (8)$$

We predict that $\bar{\sigma}_{\text{SF}}$ under housing-character weights will be at least 10% lower than under geographic weights. This is a future empirical claim, not established by the smoke package.

H2: Historic neighborhood co-assignment. Define the set of “historic tracts” as those with $\text{VT}(t) > 0.64$ (median year built before 1960) in North Carolina. We predict that the fraction of historic tract pairs that are co-assigned to the same district under housing-character weights will be at least 60%, versus at most 40% under geographic weights. The threshold remains a pre-registered target for the future state run. This tests whether the vintage signal causes the historic neighborhoods of Raleigh and Durham to cluster together rather than being split across suburban and apartment-corridor districts.

H3: Apartment corridor isolation. Define “apartment corridor tracts” as those with $MF(t) > 0.50$ (majority large-multifamily units). We predict that the mean within-district MF variance under housing-character weights is lower than under geographic weights, indicating that apartment corridor tracts are grouped together rather than scattered across single-family districts.

4.3 Package-Verified Mechanics

The smoke package `docs/examples/m-community-character-evidence-packages/M.3+acs-housing-sm` contains two synthetic tract rows and one edge. The verifier recomputes:

- `pct_single_family` from B25024_002E and B25024_003E over B25024_001E;
- `pct_multifamily` from B25024_007E–B25024_009E over B25024_001E;
- `pct_owner` from B25003_002E over B25003_001E;
- `housing_vintage` from B25035_001E with the 1940–2020 clamp;
- the final edge weight $w_h = w_{\text{geo}}(\alpha + (1 - \alpha)\text{sim}_h)$.

The package is deliberately synthetic. It validates implementation mechanics and tamper rejection; it does not claim an empirical compactness, proportionality, or community-preservation result for North Carolina.

4.4 Mill Hill Clustering — Qualitative Prediction

North Carolina has documented historic mill communities: the village neighborhoods built by textile mills in the 1910s–1940s around Durham, Burlington, and Kannapolis. Mill Hill neighborhoods share a distinctive housing character: pre-1950 construction, attached or detached single-family worker housing, moderate to high owner-occupancy. They are geographically proximate to industrial parcels (former mill sites) but have high housing character similarity to each other across municipal boundaries. We predict that under housing-character weights, Durham’s Mill Grove and West End, Burlington’s Edgewood, and similar mill villages will show elevated co-assignment rates compared to geographic weights.

4.5 Implementation Cost Confirmation

A secondary empirical goal is to confirm the zero-cost implementation claim: that adding B25024, B25003, and B25035 to the ACS fetch requires no new download infrastructure, no

new API keys, and no changes to the adjacency graph pipeline. This has been verified for the formula and graph-weight layer by:

1. extending the ACS housing CSV with the four new columns;
2. adding formula tests for denominator defaults and vintage clamping;
3. adding a hash-bound fixture that confirms the housing-character weight mode produces a positive blended edge weight.

The full all-tract North Carolina fetch and plan comparison remain future empirical work.

4.6 Comparison Methodology

All comparisons between housing-character and geographic weights will use the same seed (or convergence ensemble with $T = 600$) to isolate the effect of the weight change. The proportionality gap (D-seat share minus D-vote share) will be reported for cross-paper comparability with B-track results, with the note that proportionality is not the primary M.3 target metric.

5 Legal Grounding

5.1 Neighborhood Stability as a Community of Interest

The communities of interest doctrine, as codified in state redistricting criteria from California to Colorado to Arizona, does not enumerate the specific types of community that qualify [National Conference of State Legislatures, 2021]. California’s definition—a “contiguous population which shares common social and economic interests that would be best served by common representation” (Cal. Gov. Code §8252(c))—is functional, not categorical. Housing character satisfies this functional definition on multiple independent grounds.

Homeowner financial stake. A census tract in which 75% of units are owner-occupied is a population with a direct and legally enforceable financial stake in federal housing finance policy, historic tax credits, Community Development Block Grant allocations, and infrastructure maintenance. These federal programs affect homeowner equity directly. Renters in an adjacent apartment corridor have no such equity stake and may have opposite preferences on, for example, housing voucher policy or inclusionary zoning mandates. The owner-occupancy rate is therefore a legally cognizable marker of shared political interest, distinct from and complementary to the economic character markers in M.1.

Long-tenured homeowner representation. Courts reviewing redistricting plans have accepted testimony about the representation interests of long-established communities as evidence of community of interest. In *Harris v. Arizona Independent Redistricting Commission*, 136 S.Ct. 1301 (2016) [U.S. Supreme Court, 2016], the Supreme Court upheld the Commission’s plan against a partisan gerrymandering claim in part because the Commission had documented genuine community bonds among the affected populations. The housing vintage score operationalizes the long-tenured homeowner community: a tract with median year built before 1960 and high owner-occupancy is, by statistical inference, a community with residents who have owned their homes for decades. Their representation interests on matters of historic preservation, estate taxation, and infrastructure maintenance are shared and legally recognizable.

5.2 Historic Preservation Districts

The National Historic Preservation Act of 1966 (NHPA) established the framework for federal recognition of historic districts, codified at 16 U.S.C. §470 [U.S. Congress, 1966]. State enabling legislation in all 50 states implements parallel historic preservation programs. North Carolina’s State Historic Preservation Office maintains the National Register of Historic Places nominations for neighborhoods including Cameron Park in Raleigh, Trinity Park in Durham, and Dilworth in Charlotte.

The relevance for redistricting is twofold. First, historic district designation is an administrative recognition that a neighborhood’s built environment constitutes a community bond—precisely the kind of administrative zone co-membership that M.6 operationalizes. Second, and more directly relevant to M.3, the housing vintage score captures the underlying physical characteristic (age of construction) that qualifies a neighborhood for historic district designation. A redistricting plan that splits a pre-1940 neighborhood across district boundaries may disrupt the coordinated advocacy of historic preservation advocates for federal programs that require unified district representation.

Oregon and Colorado precedents. Oregon’s 2011 redistricting criteria, as applied by the Legislative Assembly, accepted expert testimony citing historic neighborhood cohesion as a basis for keeping Portland’s Northeast neighborhood together in a single district. Colorado’s Independent Legislative Redistricting Commission in 2021 explicitly identified “historic communities” as a recognized sub-type of community of interest under Article V, Section 44 of the state constitution [Colorado General Assembly, 2018]. The housing vintage score provides a quantitative operationalization of the “historic community” designation that is legally grounded and publicly verifiable.

5.3 Property Tax Fiscal Bond in the Housing Context

The property tax fiscal bond (M.0 and M.6) is most directly relevant to housing character in the school district context. Single-family homeowner neighborhoods are the primary funding base for local public schools: property tax revenue from owner-occupied single-family homes constitutes the majority of school district revenue in most states. A tract with high SF and high OO is a tract that is invested in local school quality in a way that a high-rental, high-MF tract is not.

Colorado’s redistricting criteria recognize this bond by listing “economic interests” and “trade area” community membership alongside traditional communities of interest [Colorado General Assembly, 2018]. The homevoter hypothesis [Fischel, 2001] formalizes the connection: homeowners in a school district are co-investors in a fiscal commons that gives them shared political interests distinct from those of renters in adjacent rental-heavy tracts.

5.4 Partisan Neutrality Analysis

Housing character weights satisfy the Shaw-Miller neutrality requirement for the same structural reasons as all M-track signals: no electoral data, voter registration data, or racial composition data enters any component of the housing character vector. The inputs are:

- ACS B25024 (housing units by structural type) — survey data, no racial or partisan content.
- ACS B25003 (tenure) — survey data, no racial or partisan content.
- ACS B25035 (median year built) — survey data, no racial or partisan content.

The correlation between housing character and race is incidental. Single-family homeowner neighborhoods are documented to be disproportionately white in many states [Rothstein, 2017], but the algorithm does not use race; it uses the housing characteristics that happen to correlate with race due to historical factors outside the algorithm’s scope. Under *Shaw v. Reno* [1993] and *Miller v. Johnson* [1995], the relevant question is whether race was the *predominant* factor. An algorithm that uses only census-bureau-published housing survey data and applies a deterministic cosine similarity formula cannot be characterized as race-predominant redistricting.

5.5 Expert Witness Deployment Template

A redistricting practitioner deploying M.3 housing character weights should structure expert testimony around the following elements:

1. Data provenance. “I used three tables from the U.S. Census Bureau’s American Community Survey 5-Year Estimates: B25024 (units in structure), B25003 (tenure), and B25035 (median year structure built), all at census tract geography. These are federal survey tables published at <https://www.census.gov/programs-surveys/acs>. No voter registration, election results, or racial composition data was used.”

2. Criterion alignment. “The [state]’s redistricting criteria require preservation of communities of interest, which I interpret to include homeowner communities with shared interests in neighborhood stability, historic preservation, and local school quality. I measured housing character as a four-component vector: the fraction of single-family units, the fraction of large-multifamily units, the owner-occupancy rate, and a vintage score reflecting the age of the housing stock. District boundaries are drawn preferentially at the boundaries between tracts with dissimilar housing character.”

3. Specific community preservation. “In the proposed plan, [specific historic neighborhood, e.g., ‘Cameron Park and Five Points in Raleigh’] are placed within a single congressional district. Both neighborhoods have median year built before 1955 and owner-occupancy rates above 65%. Their housing character similarity score is [computed from the ACS housing columns]. In the challenger’s plan, these neighborhoods are split across two districts, separating a community whose shared investment in historic preservation, property values, and school quality has been documented over decades.”

4. When not to use M.3.

- States without a communities-of-interest criterion.
- States with uniform housing stock (no meaningful within-state housing character variation).
- Contexts where housing character weights would reduce minority opportunity districts below VRA thresholds.

6 Conclusion

This paper has introduced M.3, the housing character implementation of the M-track community character framework. Three ACS tables—B25024 (units in structure), B25003 (tenure), and B25035 (median year structure built)—produce a four-component vector that captures

the single-family/multifamily dimension, the owner/renter dimension, and the historic/new-construction dimension of residential built environment character. The cosine similarity of housing character vectors modifies METIS edge weights via the M-track blend formula, causing district boundaries to fall preferentially at the transitions between single-family homeowner communities and multifamily rental corridors, and between historic pre-war neighborhoods and postwar suburban tracts.

Several features make M.3 distinctive within the M-track family.

Zero implementation cost. M.3 requires no new data download infrastructure. The three ACS tables are available from the same Census API endpoint already used by `bisect fetch`. Four new columns extend the existing demographics CSV without restructuring the pipeline. This makes M.3 the fastest M-track signal to move from specification to implementation.

Legal recognition depth. The housing character signal has unusually deep legal recognition as a community-of-interest indicator. Homeownership creates a legally enforceable fiscal bond through property taxation. Historic district designation creates an administrative recognition of community cohesion. The homevoter hypothesis [Fischel, 2001] provides theoretical grounding for why housing character predicts shared political interests. Expert testimony from redistricting proceedings in Oregon and Colorado has accepted historic neighborhood cohesion as a community of interest.

Orthogonality to existing signals. M.3 is orthogonal to M.1 (economic character): M.1 describes the workplace side of a census tract, while M.3 describes the residential built environment. A residential tract can have low economic character (few jobs) but high housing character homogeneity. M.3 is also largely orthogonal to M.6 (administrative zone co-membership), though the two signals correlate in the school district dimension. Composing M.3 and M.6 weights would capture both the physical built environment and the administrative fiscal bond dimensions of community cohesion simultaneously.

Future work. Three natural extensions follow from this paper. First, empirical validation using the implemented ACS housing pipeline will test the within-district homogeneity hypotheses (H1–H3) on North Carolina. Second, a multi-state sweep comparing housing-character weights to geographic baseline across all 50 states will identify which states have sufficient housing character variation to benefit from M.3. Third, the composition of M.3 with M.1 and M.6—a three-signal composite community character weight—is the natural

next experiment for states with strong communities-of-interest criteria and diverse built environments.

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